

BUYU DENG

School of the Gifted Young, University of Science and Technology of China
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EDUCATION

University of Science and Technology of China

B.S. Candidate in Geophysics, School of the Gifted Young
GPA: 3.62/4.30; rank: 3/20

Selected Coursework: Programming in Geophysics (94), Intro. to Seismic Exploration (98), Geophysical Data Processing and Analysis (93), Linear Algebra (90), Probability & Statistics (92), Theoretical Mechanics (95), Python & Deep Learning (A).

Hefei, China

Expected June 2027

RESEARCH INTERESTS

AI for exploration geophysics; seismic interpretation, geomodeling, rock physics, and forward modeling.

RESEARCH EXPERIENCE

Stanford Center for Earth Resources Forecasting (SCERF), Stanford University

Undergraduate Research Intern; advisor: Prof. Tapan Mukerji

Stanford, CA

July – October 2026

CO₂ Storage Modeling and Rock Physics Agent Development

- Compared five vertical grid resolutions (0.04–0.92 ft) in ECLIPSE 300 CO₂STORE simulations using fixed-time plume geometry, fixed-location saturation histories, space-time plume metrics, and computational cost; selected 0.12 ft for subsequent modeling.
- Built a facies-based rock physics workflow using mineral averaging, Hertz–Mindlin dry frames, Wood fluid mixing, and Gassmann substitution to map geology and CO₂ saturation to elastic properties.
- Generated baseline and time-lapse seismic responses for 20 years of CO₂ injection and 80 years after shut-in; interpreted horizon-relative amplitude slices in Petrel.
- Designed an agent workflow around the Stanford Rock Physics and Borehole Geophysics (SRB) model library exposed through Model Context Protocol (MCP) tools for model discovery, input validation, and traceable calculations.
- Conducted 45 independent agent runs across Smeaheia, glass-bead, and White patchy-saturation benchmarks under three resource settings: no references, textbook guidance only, and textbook guidance plus SRB tools.
- Without references, agents made unsupported model assumptions; textbook guidance matched reference calculations. SRB MCP tools enabled direct model calls and matched the references when used correctly, but some runs misinterpreted tool outputs and produced incorrect predictions.

Presentation: [Final slides \(PDF\)](#)

Computational Interpretation Group (CIG), University of Science and Technology of China

Undergraduate Researcher; advisor: Prof. Xinming Wu

Hefei, China

August 2024 – Present

Geo-VAE for 3D Geophysical Data Compression and Latent Processing

- Adapted Wan-VAE with cached causal pseudo-temporal resampling along the crossline axis, achieving isotropic $8 \times 8 \times 8$ downsampling and 31.0:1 scalar compression for 256^3 cubes.
- Demonstrated strong reconstruction fidelity across five 3D geophysical modalities while achieving a favorable balance of GPU memory use and inference time among tested state-of-the-art video VAEs on 128^3 – 512^3 cubes.
- Built a unified compact latent space for RGT-conditioned seismic generation, enabling geological structure control in the compressed domain.
- Denoised an entire 768^3 field seismic volume in a single latent-domain pass, eliminating the need for patchwise or repeated processing.

Code: github.com/BuyuDeng/Geo-VAE

RESEARCH EXPERIENCE (CONTINUED)

3D All-in-One Seismic Denoising

- Built a synthetic 3D seismic dataset with random noise, acquisition footprints, and migration artifacts; trained an all-in-one denoising network based on Degradation Representation Learning.

Code: github.com/BuyuDeng/NDR_Seismic Noise addition slides: [Download PDF](#)

ADDITIONAL RESEARCH PROJECTS

PHY-to-BGC Ocean Biogeochemical Emulation

- Built a 2.5D full-image U-Net emulator mapping global physical ocean fields to coupled biogeochemical variables, including evaluation under anomalous ocean-climate conditions. Manuscript in preparation.

GNSS–NISAR Ionospheric Super-Resolution Reconstruction

- Aligned GNSS TEC-derived phase with NISAR split-spectrum interferometric phase; tested a residual U-Net on 28 paired samples and found that sparse GNSS input preserved broad trends but did not recover NISAR-scale structure.

Report: [English project report \(PDF\)](#)

PUBLICATIONS AND PRESENTATIONS

B. Deng, Y. Dou, Z. Guo, and X. Wu. "Geo-VAE: A Unified Variational Autoencoder for 3D Geophysical Data Compression and Latent Processing." Submitted to *Geophysics*; major revision requested, 2026.

B. Deng. "Geo-VAE: A Unified Variational Autoencoder for 3D Geophysical Data Compression and Latent Processing." Oral presentation, International Meeting for Applied Geoscience & Energy (IMAGE 2026), Houston, TX, August 2026. Paper (PDF)

B. Deng. "CO₂ Storage Modeling and Rock Physics Agent Development." SCERF Seminar, Stanford University, September 2026. [Download slides \(PDF\)](#)

Y. Dou, X. Wu, Z. Guo, H. Gao, and **B. Deng**. "Implicit Structural Modeling via Generative Diffusion Frameworks." *arXiv:2606.07165*, 2026. arxiv.org/abs/2606.07165.

RESEARCH SKILLS

Python, seismic interpretation, forward modeling, rock physics, CO₂ flow simulation, ECLIPSE 300, and Petrel.

HONORS AND AWARDS

Zhao Jiuzhang Elite Program Scholarship, School of Earth and Space Sciences	2025
Rose Fund Endeavor Scholarship, School of the Gifted Young	2024
Zhao Jiuzhang Elite Program Scholarship, School of Earth and Space Sciences	2024
Freshman Scholarship, University of Science and Technology of China	2023

FIELD EXPERIENCE AND LEADERSHIP

IPGP Geophysical Field Camp, Institut de Physique du Globe de Paris August 2025
Conducted geophysical surveys and joined academic seminars with IPGP faculty.

Head (Minister), Department of Cultural & Sports, College Student Union June 2025 – January 2026